



اُونِيُورسِيْتِي تِيكْنُولُوجِي مَارَا
UNIVERSITI
TEKNOLOGI
MARA

**FACULTY OF COLLEGE OF COMPUTING, INFORMATICS AND MATHEMATICS
BACHELOR OF INFORMATION SCIENCE (HONOURS) INFORMATION
SYSTEMS MANAGEMENT**

**IMS566 ADVANCED WEB DESIGN DEVELOPMENT AND
CONTENT MANAGEMENT**

**GROUP ASSIGNMENT
ROOM BOOKING SYSTEM**

**GROUP
NCIMBSF7A**

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1.0 Introduction

1.1 Overview of the System

The **Room Booking System** is a web-based application developed using the Laravel framework to simplify the room reservation process within an organization or educational institution.

Traditionally, room reservations are completed using paper forms or manual spreadsheets. This manual process often results in duplicated bookings, misplaced records, slow approval processes, and difficulties in tracking room availability. It also requires staff to manually verify room schedules, making the process inefficient and time-consuming.

To overcome these challenges, this project converts the traditional paper-based booking form into an online booking management system. Users can register an account, browse available rooms, submit booking requests, and manage reservations through a centralized web application.

The system stores all booking information digitally in a database, allowing administrators to maintain accurate records while reducing paperwork and human errors. By automating room management and booking records, the system provides a more organized, secure, and efficient workflow.

1.2 Objectives

The objectives of the Room Booking System are:

- To convert manual room booking forms into an online system.
- To provide a centralized database for room and booking records.
- To improve the efficiency of room reservation management.

1.3 Scope

The developed system includes:

- User Registration
- User Login & Logout
- Room Management

- Booking Management
- Search Function
- Booking Report Export (PDF)
- User Profile Management
- Responsive User Interface

2.0 Discussion

2.1 GitHub Repository

GitHub Repository:

<https://github.com/nurhannanihazlan-create/room-booking-system>

Live Demonstration:

<https://room-booking-system-production-ztuk2h.laravel.cloud/>

2.2 Entity Relationship Diagram (ERD)

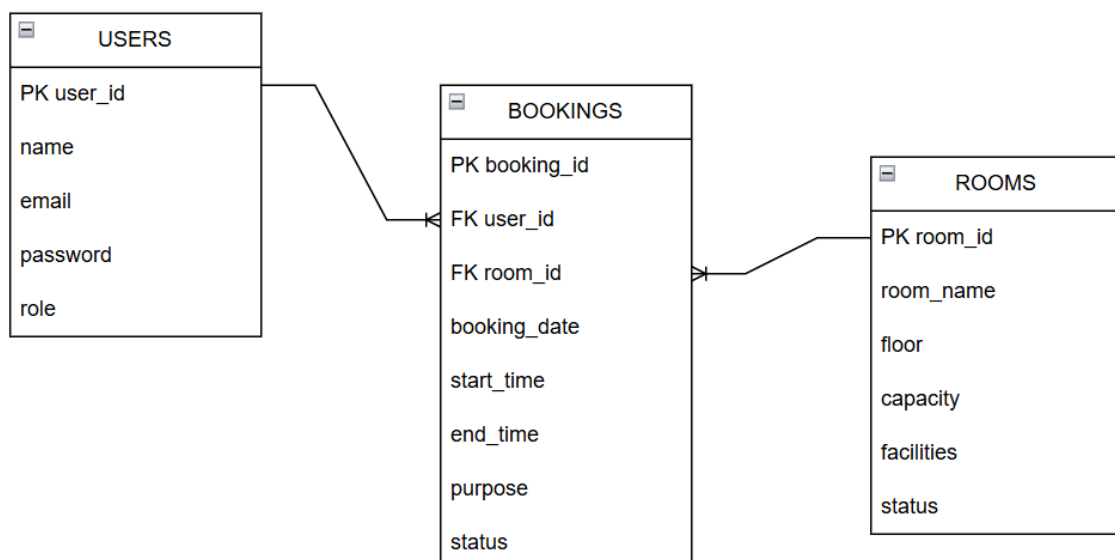


Figure 1 Entity Relationship Diagram

Database Tables

Users

Field	Type
id	Primary Key
name	varchar
email	varchar
password	varchar
timestamps	datetime

Table 1 Users Table

Rooms

Field	Type
id	Primary Key
room_name	varchar
building	varchar
floor	integer
capacity	integer
facilities	text
status	varchar
timestamps	datetime

Table 2 Rooms Table

Bookings

Field	Type
id	Primary Key
user_id	Foreign Key
room_id	Foreign Key
booking_date	date
start_time	time
end_time	time
purpose	text
status	varchar
timestamps	datetime

Table 3 Bookings Table

Relationships

- User (1) has many bookings
- Booking belongs to Room (1)

2.3 System Requirements

Software

Software	Version
Windows 10 / 11	Latest
PHP	8.3
Laravel	13
Composer	Latest
Node.js	Latest LTS
MySQL	8.0
Laragon	Latest
Visual Studio Code	Latest
Git	Latest
Google Chrome	Latest

Table 4 Software Requirements

Dependencies

- Laravel Breeze
- Tailwind CSS
- Vite
- Laravel DomPDF
- MySQL Database

2.4 User Interface Overview

The system consists of the following pages:

Home Page

Provides an introduction to the system with Login and Register buttons.

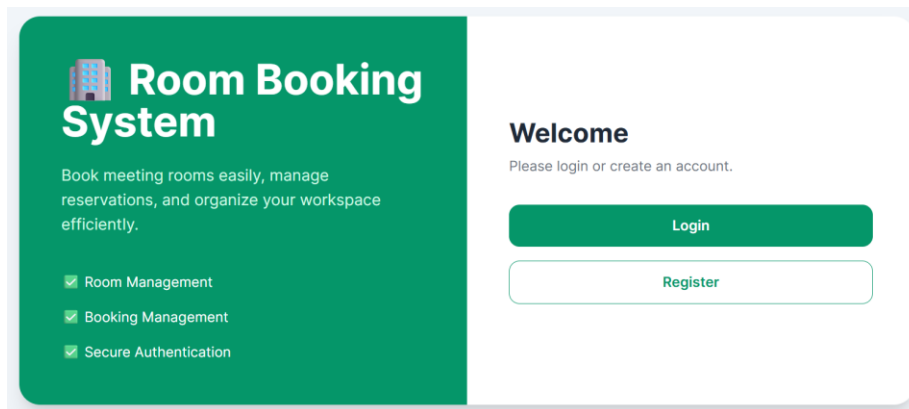


Figure 2 Home Page

Login

Allows registered users to access the system securely.

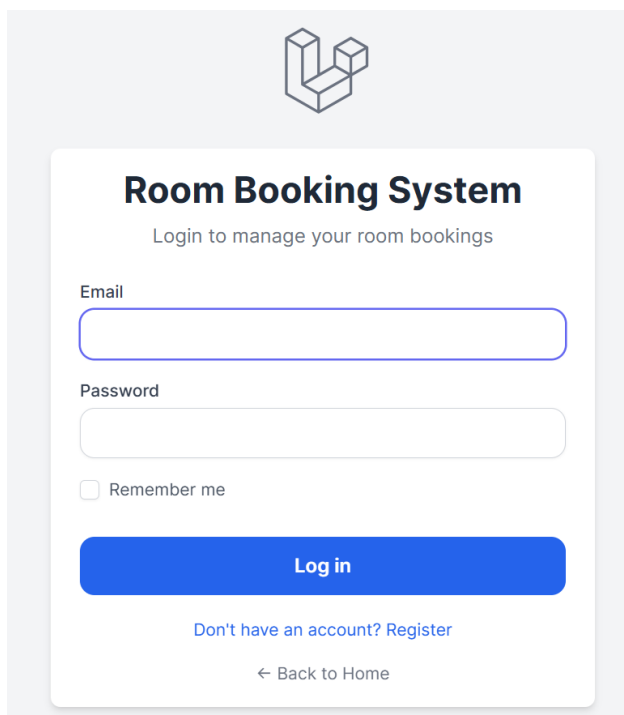


Figure 3 Login Page

Register

Allows new users to create an account.



Create Account

Register to start booking rooms

Name

Email

Password

Confirm Password

[Register](#)

[Already registered? Log in](#)

[← Back to Home](#)

Figure 4 Register Page

Dashboard

Displays a summary of:

- Total Rooms
- Available Rooms
- Total Bookings
- Pending Bookings

Also includes quick navigation to other modules.

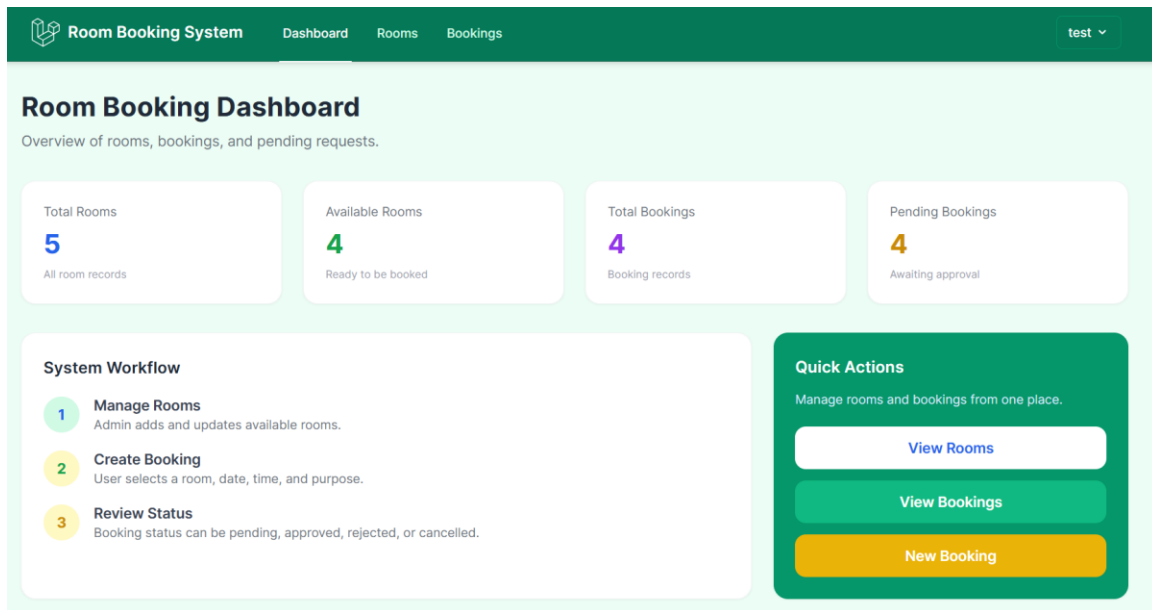


Figure 5 Dashboard Page

Room Management

Users can:

- View rooms
- Add room
- Edit room
- Delete room
- View room status



Room Booking System

Dashboard

Rooms

Bookings

test

▼



Room Management

Manage room details, capacity, and availability status.

+ Add Room

ID	Room Name	Building	Capacity	Status	Action
2	Discussion Room 1	Tower A	5	Available	Edit Delete
3	Meeting Room 1	Tower A	10	Available	Edit Delete
4	Meeting Room 2	Tower A	20	Available	Edit Delete
5	Game Room	Tower A	5	Maintenance	Edit Delete
6	Board Room	Tower A	50	Available	Edit Delete

Figure 6 Room Management Page

Booking Management

Users can:

- Create booking
- View booking list
- Search booking
- Edit booking
- Delete booking
- Export booking report


Room Booking System

Dashboard
Rooms
Bookings

test



Booking Management

View, search, update and manage all room bookings.

Export PDF

+ New Booking

Search

Reset

ID	ROOM	USER	DATE	TIME	PURPOSE	STATUS	ACTION
8	Board Room	Dr. Aminah	2026-07-12	10:00:00 - 12:00:00	Meeting with stakeholders	Pending	View Edit Delete
5	Meeting Room 2	farees	2026-07-13	11:00:00 - 12:00:00	Meeting	Pending	View Edit Delete
3	Meeting Room 1	farees	2026-07-13	11:00:00 - 12:00:00	Meeting	Pending	View Edit Delete
2	Discussion Room 1	test	2026-07-08	18:05:00 - 18:06:00	interview_edit1	Pending	View Edit Delete

Figure 7 Booking Management Page

Profile

Allows users to:

- Update personal information
- Change password
- Delete account


Room Booking System

Dashboard
Rooms
Bookings

test

Profile

Profile Information

Update your account's profile information and email address.

Name

Email

SAVE

Update Password

Ensure your account is using a long, random password to stay secure.

Current Password

Update Password

Ensure your account is using a long, random password to stay secure.

Current Password

New Password

Confirm Password

SAVE

Delete Account

Once your account is deleted, all of its resources and data will be permanently deleted. Before deleting your account, please download any data or information that you wish to retain.

DELETE ACCOUNT

Figure 8 Profile Page

2.5 Features and Functionalities

Feature	Description
User Authentication	Register, Login, Logout
Dashboard	System summary
Room Management	CRUD operations
Booking Management	CRUD operations
Search	Search booking records
PDF Export	Generate booking report
Responsive Design	Mobile and desktop compatible
User Profile	Update profile & password

Table 5 Feature and Functionalities

2.6 Workflow (CRUD Cycle)

Room Management

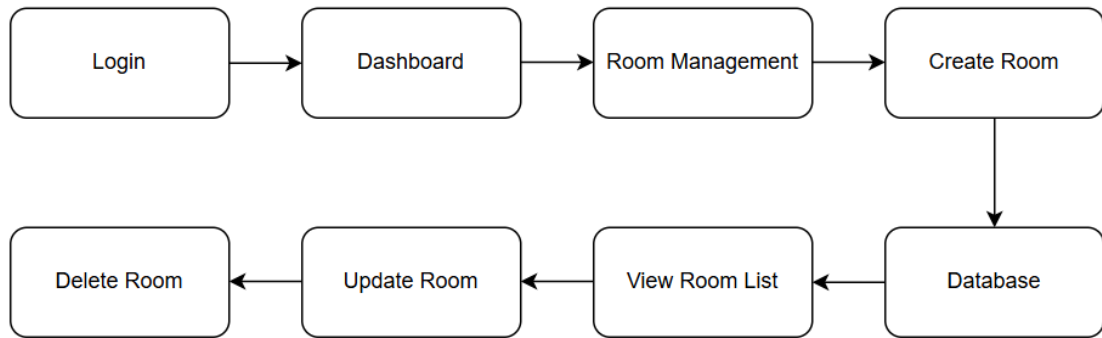


Figure 9 Room Management Workflow

Booking Management

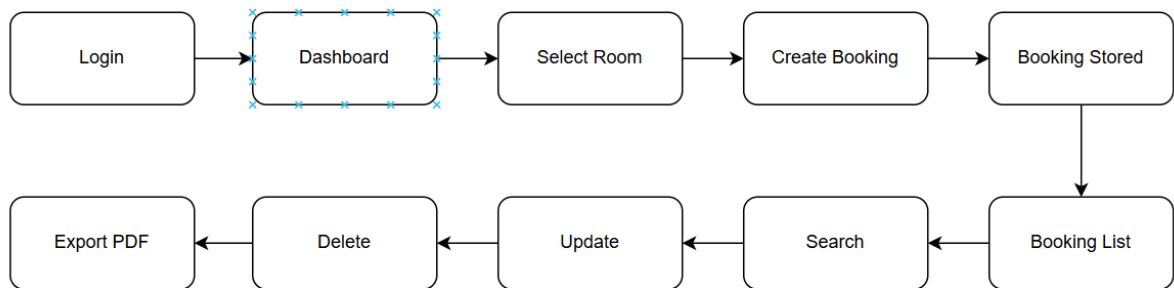


Figure 10 Booking Management Workflow

2.7 Team Roles and Contributions

Member	Role	Responsibilities
NURHANNANI IZZATI BINTI HAZLAN	Project Manager	Planning, coordination, documentation
NURHANNANI IZZATI BINTI HAZLAN	Backend Developer	Laravel controllers, models, database
NOOR ADILLA ARFINA BINTI HADY	Frontend Developer	User Interface, Tailwind CSS
SITI NAZIHAH BINTI MD SALLEH	Database Designer	ERD, MySQL, migration
SITI HAJAR BINTI ESA	Tester	Testing, debugging, report review

Table 6 Team Roles and Contributions

2.8 Contact Information

For enquiries regarding the Room Booking System:

Project Team

Faculty of Information Management

Universiti Teknologi MARA (UiTM)

Email: roombooking@uitm.edu.my

GitHub Repository:

<https://github.com/your-group-name/room-booking-system>

3.0 Conclusion & Reflection

The Room Booking System successfully transformed a manual paper-based booking process into a centralized web application using Laravel. Through the implementation of authentication, room management, booking management, PDF report generation, and responsive user interfaces, the project demonstrated the practical application of modern web development concepts.

The system improves efficiency by reducing manual paperwork, minimizing booking conflicts, and providing better organization of booking records. Users are able to access the system through an intuitive interface, while administrators can manage room information and booking records efficiently.

From this project, the team gained valuable experience in Laravel MVC architecture, database design, CRUD operations, authentication, responsive web design using Tailwind CSS, version control with GitHub, and collaborative software development. The project also strengthened problem-solving, communication, and teamwork skills throughout the software development lifecycle.

Future improvements may include an automated booking approval workflow, email notifications, room availability calendars, QR code check-ins, and role-based access control to further enhance the system's functionality and user experience.